

Lesson 14 — Sensor Thresholds & Multi-Way Decisions (+ Debug)

Matches LEGO Teacher Portal lesson: "Robot in the Deep".

Objective (I can...): I can use sensor thresholds to build a multi-way decision (more than two outcomes). **CS concept:** Thresholds; chained/else-if selection · **Builds on:** Lesson 13 (compound conditions).

AI Connection (2 min): A model sets a confidence threshold — "act only above 70%" — and can have tiers: very sure, somewhat sure, unsure. *First-principles question:* "What goes wrong if the threshold is set too low? Too high?"

Materials: LEGO CS & AI kit + device per pair, objects at varied distances / color set, "broken program" cards, notebooks. **Model reuse:** Conditionals model from Lesson 11 — code only.

60-minute flow - Warm-up (8): "Pick the cutoffs" — pairs set two distance cutoffs that split "close / medium / far" and defend them. Introduce multi-way decisions. - **Concept + demo (10):** Demo chained conditions: if very close → back up; else if medium → slow; else → cruise. Show the live value crossing each threshold. (*Rotate format: "coach ships a bug, team catches it" — a threshold set backwards.*) - **Build/Code (30):** Pairs code a three-outcome decision from sensor thresholds. **Debug task (5 min):** receive a program whose thresholds overlap or are mis-ordered so one branch is unreachable; diagnose and fix, logging the reasoning. *Swap at 15 min.* - **Challenge (7):** Tune the cutoffs and test how the behavior boundaries shift. - **Share + pack-up (5):** Pairs demo all three outcomes at three distances and explain their cutoffs. Model stays built.

Pair roles: Driver codes, Navigator moves the object and calls each threshold crossing. Swap at 15 min. (*Trio: third leads the debug.*)

Differentiation: *Support* — provide the two cutoff values. *Extension / fast-finisher* — four outcomes, or combine a threshold with a compound condition. *Reteach* — watch the live number and set one cutoff at a time, testing each.

Exit check: Pair's model reliably produces three distinct behaviors across sensor thresholds, and they fixed the unreachable-branch bug. **At-home (optional):** Describe a real "smart" behavior with three tiers (e.g., screen brightness by light level) as chained conditions. ~10 min.